

Use a separate answer sheet for each problem. Make sure to write your name, student ID and the problem number on each sheet.

1. (a) Find the solution of $Ax = b$:
$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

Solution: $x_1 = b_4/4$, $x_2 = b_2/3$, $x_3 = b_3/2$, $x_4 = b_1$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_4/4 \\ b_2/3 \\ b_3/2 \\ b_1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1/4 \\ 0 & 1/3 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

(b) Find the inverses of $A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 0 & 0 & 6 & 7 \\ 0 & 0 & 5 & 6 \end{bmatrix}$.

Solution:

$$A^{-1} = \begin{bmatrix} 0 & 0 & 0 & 1/4 \\ 0 & 0 & 1/3 & 0 \\ 0 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix},$$

$$B^{-1} = \begin{bmatrix} [3 & 2]^{-1} & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 0 & 0 & [6 & 7]^{-1} \\ 0 & 0 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 3 & -2 & 0 & 0 \\ -4 & 3 & 0 & 0 \\ 0 & 0 & 6 & -7 \\ 0 & 0 & -5 & 6 \end{bmatrix}$$

2. Tridiagonal matrices have zero entries except on the main diagonal and the two adjacent diagonals. Factor these symmetric matrices into LDL^T .

$$A = \begin{bmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 3 & 4 \end{bmatrix}$$

Solution: $E_{32}E_{21}A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{bmatrix}$

$$= \begin{bmatrix} a & a & 0 \\ 0 & b & b \\ 0 & 0 & c \end{bmatrix} = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}^T$$

$$E_{43}E_{32}E_{21}B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}^T$$

3. Suppose $Q^T = Q^{-1}$ (transpose equals inverse, so $Q^T Q = I$).
- (a) Show that the columns $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n$ are unit vectors: $\|\mathbf{q}_i\|^2 = 1$.
- (b) Show that every two columns of Q are perpendicular: $\mathbf{q}_i^T \mathbf{q}_j = 0$ if $i \neq j$.
- (c) Find a 2 by 2 example with first entry $q_{11} = \cos \theta$

Solution:

$$\text{Since } Q = [\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n], \quad Q^T = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_n^T \end{bmatrix}, \quad Q^T Q = \begin{bmatrix} \mathbf{q}_1^T \mathbf{q}_1 & \cdots & \mathbf{q}_1^T \mathbf{q}_n \\ \vdots & \ddots & \vdots \\ \mathbf{q}_n^T \mathbf{q}_1 & \cdots & \mathbf{q}_n^T \mathbf{q}_n \end{bmatrix} = I$$

or $[Q^T Q]_{ij} = \mathbf{q}_i^T \mathbf{q}_j = \delta_{ij}$, therefore

(a) $\mathbf{q}_i^T \mathbf{q}_i = \|\mathbf{q}_i\|^2 = 1,$

(b) $\mathbf{q}_i^T \mathbf{q}_j = 0$ if $i \neq j;$

(c) $Q = \begin{bmatrix} \cos \theta & \mp \sin \theta \\ \pm \sin \theta & \cos \theta \end{bmatrix}$

4. There are three lines on the xy plane
- $L1: x + y = 1$
 $L2: x + 2y = 2$
 $L3: 2x + y = 2$

(a) Find the solution of $L1$ and $L2$ as $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Solution:

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- (b) Find the point $P(x_0, y_0)$, which minimize the function
 $f(x, y) = (x + y - 1)^2 + (x + 2y - 2)^2 + (2x + y - 2)^2$.
(hint: $A^T A x = A^T b$)

Solution:

$$\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A^T A = \begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix}, \quad A^T \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix} \rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 6 & -5 \\ -5 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 7 \end{bmatrix} = \frac{7}{11} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$